

MANMA

RAG Knowledge Base

Mayank Manohar / MAYANK.OS

A structured knowledge document for the portfolio AI assistant **Manma**. It contains the established portfolio, skills, projects, learning direction, and website context available from the user's prior work.

Primary purpose: document ingestion → chunking → embeddings → vector search → grounded Gemini responses.

Important: Manma should not invent facts that are not present in this knowledge base.

RAG-ready structure

- Profile and professional identity
- Education and career direction
- Technical skills and tools
- Projects and portfolio work
- AI / LLM learning direction
- MAYANK.OS portfolio experience
- Manma assistant purpose and behavior
- Contact and public-profile placeholders where exact values are not available

1. Professional Profile

Name: Mayank Manohar

Professional identity: Full Stack Developer

Primary development focus: MERN stack, backend systems, deployment, and applied AI.

Current positioning: Mayank is building toward software engineering roles where full-stack development and useful AI applications meet. He enjoys taking an idea from interface to API, data model, deployment, and usable product.

Current AI direction: Mayank is exploring LLM APIs, LangChain, LangGraph concepts, embeddings, vector databases, RAG, and AI assistants.

Portfolio identity: The personal portfolio is branded as **MAYANK.OS**. The site is designed as an interactive, animated developer portfolio rather than a plain resume page.

2. Education & Career Context

College: Dr. B. C. Roy Engineering College

Degree: B.Tech engineering

Expected passing year: 2027

Location used in internship context: Durgapur

Career goal: prepare for off-campus software engineering opportunities, with an emphasis on larger-company roles rather than a TCS-style career path.

Placement preparation: Mayank has been working on DSA/coding preparation, GitHub, LeetCode, LinkedIn, projects, deployment, and off-campus applications.

3. Technical Skills

Frontend: React.js, JavaScript, HTML, CSS, Tailwind CSS, GSAP

Backend: Node.js, Express.js, REST APIs

Databases: MongoDB, SQL

DevOps / deployment: Docker, AWS concepts and deployment workflows

Version control: Git, GitHub

Programming: JavaScript, Java, Python

AI / GenAI: LLM APIs, LangChain, embeddings, vector databases, RAG, and AI assistant development.

Current Manma backend stack: Node.js + Express.js + LangChain.js + Gemini + vector database. Mayank chose JavaScript because he is already comfortable with the MERN ecosystem.

Portfolio animation: Lenis has been added for smooth scrolling. GSAP and ScrollTrigger are considered/used for scroll-triggered animation and interaction.

4. AI / GenAI Direction

Mayank is learning how to build practical AI applications rather than only studying AI theory. The main concepts he is interested in are LLM APIs, prompt design, embeddings, vector databases, semantic retrieval, RAG, and AI

agents/workflows.

Manma: Manma is the portfolio's AI assistant. Its job is to help visitors understand Mayank quickly.

Manma use cases: explain who Mayank is, explain his skills, explain projects, answer questions about his technical direction, surface relevant project links, and guide visitors toward contact information.

RAG principle: Manma should retrieve relevant information from Mayank's curated knowledge base before generating an answer. If the knowledge base does not contain the answer, Manma should clearly say that the information is unavailable rather than inventing it.

Current API direction: the portfolio communicates with the Manma backend through a POST endpoint at `/manma`.

Current model setup: Gemini is being used successfully with LangChain. The exact model identifier should be treated as the value configured in the live project rather than hard-coded into this document.

5. Portfolio Projects

RasoiMitra — Smart Recipe Companion

RasoiMitra is a modern interactive web application for discovering recipes, viewing nutrition information, following step-by-step cooking instructions, and accessing video tutorials.

Stack: HTML, CSS, JavaScript, Node.js backend, Spoonacular API.

Features: ingredient-based recipe discovery, step-by-step instructions, nutrition information, video tutorials, animated welcome experience with floating food emojis, responsive design, hidden scrollbars, and a sticky back button.

Deployment work: The project was pushed to Mayank's GitHub after working through repository and deployment issues. A Vercel CSS/image loading problem was resolved by correcting case-sensitive asset paths.

Current status: The project was described as working with its intended functions and is suitable for inclusion in the portfolio/resume.

Employee Payroll System

A Java desktop application connected to MySQL for employee and payroll management.

Technology: Java, MySQL, Swing, JFreeChart.

Features: employee CRUD operations, payroll management, payroll/history views, department visualization, CSV-related functionality, payslip functionality, leave deduction, and hike/salary-related functionality.

Database: MySQL database context included a database named `payroll_db`.

SkyCast Weather App

A weather application built using HTML, CSS, and JavaScript with the OpenWeatherMap API.

Phishing Classifier

A machine-learning project for checking whether a given website is likely to be a phishing website or a legitimate website.

Technology: Python, NumPy, Pandas and machine-learning techniques.

Project date in resume history: February 2025.

Live Selling Analysis Project

A data-analysis project using Python, Pandas, Matplotlib and scikit-learn. The project involved imputation, outlier detection, exploratory analysis of reactions/comments/shares, and clustering.

Project date in resume history: June 2024.

Quest to Find Efficient Option Trading Algorithm

A project listed in Mayank's earlier resume material. The historical project description included machine-learning classification and clustering work, including hyper/hypothyroid-related classification. Because the available historical description was inconsistent with the project title, Manma should not invent or over-explain this project's exact objective without a newer verified project description.

Project date in resume history: December 2024.

Logiphex UI/UX website

Mayank built a website associated with the Logiphex authentication page and worked on UI/UX. The available project record does not contain enough verified technical detail to describe its backend or exact feature set. Manma should keep the description general unless the portfolio knowledge base is updated.

6. MAYANK.OS Portfolio

Brand: MAYANK.OS. The '.os' identity was selected because it fits the systems-oriented visual direction of the portfolio better than a conventional '.dev' presentation.

Visual direction: modern editorial developer portfolio with a dark/light theme, lime accent, large typography, thin borders, structured sections, system-like labels, and subtle motion.

Hero concept: the hero communicates Mayank as a full-stack developer focused on useful interfaces, backend systems, deployment, and applied AI. The right-side visual is an abstract 'live build environment' orbital/system graphic representing interface, API/data, and AI exploration layers.

Profile section: explains Mayank's full-stack background and systems mindset. A professional photo or avatar can be used here so that the human identity remains distinct from Manma.

Skills section: should use recognizable technology icons in a clean grid rather than progress bars. The intended presentation is icon + technology name, with consistent sizing and subtle hover/scroll animation.

Projects section: should support many projects without becoming a long stack of cards. Filtering/sorting by categories such as Full Stack, AI/ML, Frontend, and Backend should remain usable, with an animated/scrolling presentation.

Contact: direct email should open Gmail, and the visible contact number can be used as a direct contact CTA. Exact phone number and email values are not included in this knowledge document because they were not available as verified values in the retrieved context.

Command palette: the portfolio includes a keyboard command palette concept. It should provide useful navigation/actions such as opening Manma, going to Work/About/Stack/Contact, and opening external profile links.

Animation: Lenis is used for smooth scrolling; GSAP/ScrollTrigger can provide scroll-triggered reveals, project transitions, Manma drawer transitions, and subtle interaction effects.

7. Public Profile & Career Branding

GitHub: username/handle used in profile work: **manohar-mayank**.

LinkedIn positioning: Full Stack Developer / MERN Stack with Docker and AWS, with React.js, Node.js, MongoDB, and Express.js highlighted.

Resume direction: recruiter-facing wording should emphasize practical full-stack development, deployment, projects, and applied AI rather than listing technologies without evidence.

Google Student Ambassador: Mayank was selected at Dr. B. C. Roy Engineering College and completed assigned tasks/campus activities. The experience included Google-related campus events and a completion certificate.

8. Learning & Engineering Context

Mayank is comfortable with MERN and JavaScript-based backend development. He has been learning deployment and cloud concepts including Docker, AWS EC2/ECR, VPCs, subnets, security groups, load balancers/target groups, and backend deployment workflows.

Mayank has explored microservice architecture for backend deployment and considered Nginx as part of deployment architecture. These are learning areas; Manma should distinguish them from technologies used in a completed project.

Mayank has worked with Node.js version 22.14.0 in his development environment and commonly uses Windows + VS Code. This environment detail is mainly useful for technical/project context and should not be volunteered unless relevant.

For AI portfolio work, Mayank prefers staying close to the MERN/JavaScript ecosystem when possible rather than introducing a Python framework unnecessarily.

9. Manma: Assistant Identity & Behavior

Name: Manma

Role: Personal AI guide for the MAYANK.OS portfolio.

Purpose: help a visitor understand Mayank's background, projects, technical stack, learning direction, and ways to contact him without forcing the visitor to read the entire portfolio.

Tone: concise, friendly, technically clear, professional, and natural.

Grounding rule: answers should be based on the retrieved portfolio knowledge. Do not fabricate projects, technologies, achievements, dates, links, employers, clients, or personal information.

Unknown-information rule: if the knowledge base does not contain the answer, say that Manma does not have that information yet and, when appropriate, direct the visitor to the relevant portfolio/contact section.

Project linking: when a question clearly relates to a project, Manma should return the project's relevant title and available live/GitHub link from metadata. Links should come from the project data, not be invented.

Source awareness: Manma can mention the relevant source area, such as About, Skills, or a project document, when the UI supports source/context indicators.

10. Example Visitor Questions

Q: Who is Mayank?

Manma: Mayank Manohar is a Full Stack Developer and B.Tech engineering student building toward software engineering roles. He works mainly with the MERN stack and is exploring applied AI.

Q: What is Mayank's tech stack?

Manma: His main stack includes React.js, JavaScript, Node.js, Express.js, MongoDB, SQL, Docker, AWS concepts, Git/GitHub, and GSAP. He is also exploring LLM APIs, LangChain, embeddings, vector databases, and RAG.

Q: Tell me about RasoiMitra.

Manma: RasoiMitra is a smart recipe companion built with HTML, CSS, JavaScript, a Node.js backend, and the Spoonacular API. It supports ingredient-based discovery, nutrition information, instructions, and video tutorials.

Q: What AI technologies is Mayank learning?

Manma: Mayank is exploring LLM APIs, LangChain, embeddings, vector databases, RAG, and AI assistants such as Manma.

Q: What is Manma?

Manma: Manma is the AI guide built into MAYANK.OS. It helps visitors explore Mayank's profile, skills, projects, and technical direction using a grounded knowledge base.

Q: How can I contact Mayank?

Manma: Use the Contact section of MAYANK.OS for Mayank's direct email, phone, GitHub, LinkedIn, and resume. Exact contact values should be populated from the live portfolio data.

Q: What does Mayank use for backend development?

Manma: Mayank primarily uses Node.js and Express.js, with MongoDB/SQL for data and Docker/AWS concepts for deployment.

11. Suggested RAG Metadata

When ingesting these documents, attach metadata so Manma can return useful source context and project links.

- **source_type:** about | skills | education | project | experience | contact | portfolio
- **title:** human-readable document title
- **section:** logical portfolio section
- **project:** project name when applicable
- **technology:** relevant technology tags when applicable
- **url:** verified live/GitHub/project URL when available
- **updated_at:** date/version when the knowledge document is updated

12. RAG Ingestion Notes

This document is intentionally organized into sections that can be split into smaller chunks. For better retrieval, project descriptions should remain grouped with their project title and technology list.

Avoid making one giant chunk containing the entire portfolio. A visitor asking about RasoiMitra should retrieve RasoiMitra-specific content rather than unrelated education or contact information.

A practical starting point is to maintain separate Markdown source files for each major topic/project, then load and split them during ingestion. Keep the PDF as a master reference; the Markdown/JSON files can be the canonical runtime knowledge source.

13. Values to Fill From the Live Portfolio

The following should be populated from the actual portfolio/backend configuration before production use:

- Verified personal email address
- Verified phone number
- Live portfolio URL
- Verified GitHub profile URL
- Verified LinkedIn profile URL
- Live URLs and GitHub URLs for each project
- Exact current Gemini model identifier configured in the backend
- Any new projects/skills added after this document was created

RAG principle for Manma: Retrieve first, answer second. The model should use the retrieved portfolio facts as its source of truth and avoid unsupported claims.